

Process Characterization Report

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCR-008

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC — antibody-dependent cell-mediated cytotoxicity; AEX — anion-exchange chromatography; AMV — analytical-method validation; ANOVA — analysis of variance; BLA — Biologics License Application; CCD — central-composite design; Cpk — process-capability index (one-sided); CPP — critical process parameter; CQA — critical quality attribute; %CV — percent coefficient of variation; DNA — deoxyribonucleic acid; DoE — design of experiments; DS — drug substance; ELISA — enzyme-linked immunosorbent assay; HCP — host-cell protein; icIEF — imaged capillary isoelectric focusing; IgG1 — immunoglobulin G subclass 1; LRF — log reduction factor; MSAT — Manufacturing Science and Technology; MVM — minute virus of mice (parvovirus); NOR — normal operating range; PAR — proven acceptable range; PCP / PCR — Process Characterization Plan / Report; QbD — quality by design; RSM — response-surface methodology; SDM — scale-down model; SEC — size-exclusion chromatography; SOP — standard operating procedure; TCID₅₀ — 50% tissue-culture infectious dose; WC-CPP — well-controlled critical process parameter; XMuLV — xenotropic murine leukaemia virus (enveloped model virus).

Executive summary

This report documents the Stage 1 (Process Design) characterization of the A-Mab anion-exchange (AEX) polishing step, the flow-through operation that follows cation-exchange polishing and is the final chromatographic step of the drug-substance purification train. The study was executed against the paired plan PCP-008 on a qualified commercial-equivalent scale-down model, under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), the enhanced quality-by-design paradigm (International Council for Harmonisation 2009, 2023b, 2012), and — because the step carries part of the viral-safety claim — the viral-clearance evaluation framework of ICH Q5A(R2) (International Council for Harmonisation 2023a), all grounded in the A-Mab case study (CMC Biotech Working Group 2009). Anion exchange is the step at which the process establishes the minute-virus-of-mice (MVM, parvovirus) clearance claim — the only CQA it sets — while providing orthogonal enveloped-virus (XMuLV) clearance and a further reduction of host-cell protein (HCP), residual DNA and leached Protein A.

Five process parameters were characterized. Load pH, equilibration/wash-1 conductivity, load conductivity and protein load were studied together in a 19-run two-level full-factorial screening design and a 28-run face-centred central-composite response-surface design; operating flow rate was assessed univariately. The screening design identified the significant factors and two-factor interactions; the response-surface design then quantified main effects, interactions and curvature and defined the multivariate operating region. All four multivariate factors and the flow rate are classified as well-controlled critical process parameters (WC-CPP); the step has no key process parameter and no parameter requiring the most stringent (full CPP) control.

The MVM and XMuLV clearance surfaces and the pool-HCP surface are governed by the electrostatic environment: MVM log-reduction rises with load pH (effect $1.22 \log$, $p < 0.001$) and falls with load conductivity, and the pool-HCP level is set by load pH and wash-1 conductivity acting in opposition. The response-surface models are adequate over the characterized region (no significant lack of fit); the pool-HCP and XMuLV models are additionally predictive (predicted $R^2 = 0.62$), while the MVM model is adequate ($R^2 = 0.90$) with a more modest predicted R^2 that reflects the narrower log-reduction range of the parvovirus spiking assay.

Anion exchange contributes approximately $4.71 \log$ of MVM clearance at the nominal condition — the first and principal MVM-clearance step — so that the drug substance meets its $8.6 \log$ cumulative MVM requirement (achieved cumulative $10.0 \log$, $Cpk = 1.51$). This MVM capability is the tightest of the entire drug substance (process-minimum $Cpk = 1.51$), yet retains margin above the requirement. The step contributes $5.95 \log$ to the cumulative XMuLV claim ($Cpk = 1.53$) alongside low-pH inactivation and virus filtration, and holds pool HCP well within its 100 ng/mg limit ($Cpk = 6.14$). The step does not, by itself, deliver the cumulative viral-safety claim: MVM safety depends jointly on this step and virus filtration (PCR-009), and XMuLV safety on low-pH inactivation (PCR-006) and virus filtration.

Three deviations were recorded, investigated and impact-assessed. The most consequential — a non-representative, deamidated load in the first DoE execution — invalidated that execution for operating-

region definition; the affected designs were re-executed in full on a requalified, representative load, and the analysis reported here is that requalified execution. None of the three deviations altered the parameter classifications or the operating region defined below. The step is well understood and robust across the characterized ranges, and the parameter classifications, operating region and control contributions defined here are carried forward into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cell-mediated cytotoxicity; it is manufactured by the platform drug-substance train summarized in Table 1.1 and purified by the sequence Protein A → low-pH viral inactivation → cation exchange → anion exchange → virus filtration → UF/DF (CMC Biotech Working Group 2009). Anion exchange (Step 8) is the final polishing chromatography of that train.

Table 1.1: A-Mab drug-substance process train (Steps 3–10) and each step’s principal control-strategy role.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The step is operated in flow-through mode: the cation-exchange eluate is titrated and conditioned to the target load pH and conductivity and passed over a strong anion-exchange resin under conditions at which the antibody carries a slight net positive charge and does not bind, while the more acidic host-cell protein, DNA, leached Protein A and model viruses are retained. The product is therefore recovered in the flow-through and wash and the impurities and viruses are removed by binding — a mechanism that makes the step a high-recovery polish rather than a yield-limiting capture. Anion exchange forms no product quality attribute; it sets one CQA of its own, the cumulative MVM (parvovirus) viral-clearance claim, and contributes clearance of enveloped virus and of the process impurities that are formed upstream. This division of labour — one CQA set, several cleared — is

developed in Section 2 and is the reason the step is characterized against both a viral-clearance criterion and the drug-substance impurity specifications.

1.2 Objectives

The characterization had five objectives, mirroring the paired plan PCP-008: (i) to quantify the relationships between the anion-exchange process parameters and the step’s viral-clearance, impurity-clearance and yield responses; (ii) to define the multivariate operating region over which the governed attributes remain within acceptance; (iii) to establish the proven acceptable range of each parameter and locate the normal operating range within it; (iv) to classify each parameter according to its demonstrated criticality; and (v) to justify the Stage-2 operating ranges and the step’s contribution to the overall control strategy for transfer to commercial manufacture.

1.3 Regulatory and scientific basis

The study follows the lifecycle approach to process validation, in which Stage-1 process design establishes the process understanding on which the later qualification and continued verification stages depend (U.S. Food and Drug Administration 2011); the report is organized around the Stage-1 (Process Design) study record that guidance envisages. Criticality is treated as a continuum rather than a binary (International Council for Harmonisation 2023b): a parameter’s classification reflects the magnitude of its demonstrated effect on the CQAs and the width of the range over which that effect keeps the attribute in specification, consistent with the enhanced, quality-by-design development approach (International Council for Harmonisation 2009, 2012). Because anion exchange is one of three orthogonal viral-clearance steps, its clearance data are additionally evaluated within the ICH Q5A(R2) framework (International Council for Harmonisation 2023a), and its contribution is credited modularly rather than claimed in full (Section 10). The controlled documents that govern the study and its sibling documents are listed in Table 1.2.

Table 1.2: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Flow-through anion-exchange polishing is a platform operation whose behaviour is well established from related humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and from A-Mab clinical and engineering experience (CMC Biotech Working Group 2009); the characterization therefore confirms and bounds a known mechanism rather than discovering a new one. Prior knowledge established that, in flow-through mode, the binding of impurities and virus is governed by the electrostatic environment. A higher load pH increases the net negative charge of the acidic impurities and of the model viruses, strengthens their retention on the positively charged resin and therefore improves their clearance, while raising the equilibration/wash-1 or the load conductivity screens that electrostatic attraction and weakens retention. The antibody itself, at a pH below its isoelectric point, remains weakly positive and transmits across the studied range, so the step is expected to be high-recovery and largely yield-insensitive to the chromatographic factors. These expectations set the direction of every factor before the study and are the hypotheses the results confirm or refute.

2.2 Quality attributes in scope

The attributes governed by anion exchange are listed in Table 2.1: the one CQA the step sets and the four it clears, each with its drug-substance acceptance criterion and criticality. The step sets the cumulative MVM clearance — a very-high-criticality viral-safety attribute — and clears the cumulative XMuLV clearance (also very high criticality) together with three process impurities of lower criticality (HCP, residual DNA and leached Protein A). Criticality here follows the case-study risk tool: the two viral-safety attributes carry the highest severity, which is why the viral responses, not the impurity responses, are the binding constraints on the operating region and on capability.

Table 2.1: Quality attributes governed by anion exchange — one set (MVM) and four cleared.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

2.3 Risk-based prioritization and parameter selection

The parameters carried into this study, and the ranges over which they were varied, were set by the pre-characterization risk assessment RA-001, which risk-ranked the anion-exchange parameters and filtered them to those with a credible impact on the governed attributes. Four parameters — load pH, equilibration/wash-1 conductivity, load conductivity and protein load — act directly on the electrostatic clearance mechanism and were therefore studied jointly in the multivariate design, so that their interactions could be resolved rather than assumed absent. Each factor’s range was set wider than its intended normal operating range to characterize the knowledge space around the set-point: the load-pH range brackets the titration tolerance of the conditioned load, the two conductivity ranges bracket buffer-preparation and load-conditioning variability, and the protein-load range extends to a resin loading well above the intended set-point to probe the mass-transfer limit of impurity and virus binding. Operating flow rate was judged to act on clearance only through residence time and not through the binding chemistry, and was therefore assessed univariately rather than consuming a factor in the multivariate design. Consistent with the conservative default for a very-high-criticality attribute, the viral responses were characterized across the full factor ranges even where prior knowledge predicted robustness.

3 Materials and methods

3.1 Scale-down model and its qualification

The characterization was performed on a scale-down model of the anion-exchange step operated at commercial-equivalent conditions — the same resin chemistry, bed height, load ratio and linear flow velocity as the intended commercial operation — in accordance with the step operation procedure (SOP-2011) and the scale-down-model qualification procedure (SOP-1001). The model was qualified by matching its input and output attributes to at-scale development data across the operating point: the flow-through-pool impurity levels, the step yield and the viral log-reduction factors of the qualified model reproduced the at-scale values within the analytical variability of the respective methods, establishing that the model is representative for both operating-range and clearance claims. For the viral responses the model is a qualified small-scale spiking model, in which a defined titre of XMuLV or MVM is spiked into the conditioned load and the log-reduction factor is measured across the step under the ICH Q5A(R2) expectation that small-scale clearance data support the cumulative claim (International Council for Harmonisation 2023a). The replicated centre point of each design provides the pure-error estimate against which lack of fit is tested (Section 5); its reproducibility is reported in Table 5.1.

3.2 Operation

Each run conditioned the cation-exchange eluate to its target load pH and load conductivity, equilibrated the column and performed the post-load wash at the target wash-1 conductivity, and collected the flow-through and wash as the product pool under UV-triggered pool-collection control. Inputs held to platform specification — resin lot and packing quality, column geometry, buffer identities and the sanitization regime (SOP-2008) — were identity- and quality-controlled rather than characterized, so that the measured responses reflect the studied factors and not uncontrolled operational variation. The single exception is the charge-variant quality of the load material, which is a feed-input attribute that proved consequential to the study and is treated explicitly under Deviations (Section 13) and in the control strategy (Section 10).

3.3 Analytical methods

The validated methods used to measure the step’s responses are listed in Table 3.1. Pool HCP was measured by ELISA (AMV-3012), residual DNA by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016); the enveloped- and small-virus log-reduction factors were measured by the XMuLV and MVM infectivity assays (AMV-3017 and AMV-3018); and the charge-variant quality of the load, central to the load-material deviation, was measured by imaged capillary isoelectric

focusing (AMV-3013). Method performance is not reproduced here; the reportable ranges and techniques are summarized in Appendix D.

Table 3.1: Controlled procedures and validated analytical methods for the anion-exchange characterization.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID ₅₀)	Method validation
AMV-3018	MVM Infectivity Titre (TCID ₅₀ /qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

3.4 Sampling plan

Each design run yielded one flow-through pool, sampled for HCP, residual DNA, leached Protein A and step yield, with viral log-reduction determined on the spiked runs. The screening design carried 3 replicated centre points and the response-surface design carried 4, providing the independent pure-error replication used for the lack-of-fit test. Load-material charge-variant quality was verified on each load lot by icIEF prior to execution, the control that ultimately detected the non-representative load in the first execution.

3.5 Statistical methods

Factors were analysed in coded units (set-point = 0, characterization edges = -1 and $+1$). The screening data were fitted by ordinary least squares with all main effects and two-factor interactions, each effect estimate being twice the corresponding coded coefficient; the response-surface data were fitted with the full quadratic model. Significance was assessed at $\alpha = 0.05$. Model adequacy was judged by the coefficient of determination and its adjusted and predicted (PRESS-based) forms, the overall F-test, and an ANOVA partition of the residual into lack-of-fit and pure-error components using the replicated centre points; a response with no statistically active factor is reported as robust rather than modelled. Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with each parameter drawn across its normal operating range and propagated through the fitted response-surface models, from which the one-sided process-capability index (Cpk) of each governed attribute was computed.

4 Study design

4.1 Factors, ranges and the knowledge space

The five characterized parameters, their set-points, normal operating ranges, characterization ranges and final classifications are given in Table 4.1, and the coded legend for the four multivariate factors in Table 4.2. Each characterization range extends beyond the normal operating range so that the knowledge space is bounded on both sides of the set-point: load pH was varied over 7.2–7.8 around a 7.5 set-point, equilibration/wash-1 conductivity over 1.6–3.6 mS/cm, load conductivity over 3–8 mS/cm and protein load over 50–300 g/L resin. The range column is the characterization range and not a proven acceptable range; the proven acceptable ranges are computed per attribute in Section 7.

Table 4.1: Anion-exchange parameters — set-point, normal operating range, characterization range, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

Table 4.2: Coded-factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

4.2 Screening design

The four multivariate factors were first studied in a two-level full-factorial screening design of 19 runs, including 3 replicated centre points, resolving all four main effects and all six two-factor interactions clear of one another. The screening design's role is to identify which factors and interactions are active; the magnitudes it returns are refined by the subsequent response-surface model. The full coded design matrix and responses are in Appendix A.

4.3 Response-surface design

The same four factors were then studied in a face-centred central-composite design of 28 runs — the factorial points, the axial (face) points that add curvature, and 4 replicated centre points — fitted with the full quadratic model. This response-surface model is the predictive, design-space-defining model of the study: it quantifies the main effects, two-factor interactions and curvature and is the basis for the operating region (Section 6) and the proven acceptable ranges (Section 7). The full coded matrix and responses are in Appendix B.

4.4 Univariate assessment

Operating flow rate was assessed univariately across 150–450 cm/hr about its 300 cm/hr set-point rather than in the multivariate design, because it influences clearance only through residence time and not through the binding chemistry that couples the other four factors; treating it separately avoids inflating the multivariate design without loss of information. Its assessment confirmed that clearance and recovery are insensitive to flow rate over the studied range, and it is classified from that univariate evidence in Section 9.

5 Results

5.1 Centre-point performance and reproducibility

The replicated centre point establishes the pure-error baseline for the study, and its reproducibility is attribute-dependent (Table 5.1). The viral log-reduction factors and the step yield are highly reproducible at the centre point (%CV of 6.7% for XMuLV, 6.9% for MVM and 1.3% for yield), whereas the pool HCP is the least precise response (%CV 18.2%), consistent with the near-baseline HCP levels this polishing step delivers and with the ELISA’s variability at low analyte concentration. This higher HCP variance is carried through to the lack-of-fit test and is the reason HCP effects near the assay’s precision are hedged below.

Table 5.1: Centre-point reproducibility of the response-surface design (pure-error basis).

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMuLV LRF (log)	4	6.63	0.445	6.71
MVM LRF (log)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

5.2 Screening: factor effects

The screening design resolved a small number of large, mechanistically coherent effects and found the remaining terms inactive (Table 5.2); the effect estimates for pool HCP are detailed in Table 5.3, and the viral and yield responses are described below with their full effect tables in Appendix C. Two factors dominate every governed response, and they are the two the mechanism predicts.

Table 5.2: Screening-model adequacy across the four responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45
XMuLV LRF (log)	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF (log)	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

The pool HCP is governed by load pH and equilibration/wash-1 conductivity acting in opposite directions: a higher wash-1 conductivity raises the pool HCP (effect 12.1 ng/mg, $p < 0.001$, poorer

clearance) and a higher load pH lowers it (effect -11.9 ng/mg, $p < 0.001$, better clearance), with a significant load-pH $\times$ wash-1-conductivity interaction (effect -6.6 ng/mg, $p < 0.001$) that means conductivity matters most at low load pH. Notably, the protein load has no significant effect on pool HCP in this representative-load design (effect 0.89 ng/mg, $p = 0.491$), and its interaction with wash-1 conductivity is likewise absent (effect 0.22 ng/mg, $p = 0.860$) — a result that becomes central to the load-material deviation in Section 13, because that same interaction was large and significant in the invalidated first execution.

Table 5.3: Screening effect estimates for pool HCP (requalified-load execution).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

The viral responses share a single governing direction. Both the MVM and XMuLV log-reductions rise with load pH (MVM effect 1.22 log , $p < 0.001$; XMuLV effect 1.00 log , $p < 0.001$) and fall with load conductivity (MVM effect -0.71 log , $p < 0.001$; XMuLV effect -0.60 log , $p = 0.015$), exactly as the electrostatic mechanism predicts — stronger binding of the negatively charged virus at higher pH and lower ionic strength. The step yield shows no significant dependence on any factor, consistent with a high-recovery flow-through operation in which the antibody transmits regardless of the chromatographic condition. These screening magnitudes are indicative; the design-space magnitudes are taken from the response-surface models.

5.3 Response-surface models

The response-surface models are adequate across the characterized region and, for the two principal clearance responses, additionally predictive (Table 5.4). The pool-HCP and XMuLV models carry predicted R^2 of 0.66 and 0.62 respectively, so they predict about as well as they fit; the MVM model is adequate ($R^2 = 0.90$, adjusted $R^2 = 0.79$, no significant lack of fit) with a more modest predicted R^2 (0.39) that reflects the narrower log-reduction range of the MVM spiking assay relative to its reproducibility, and it is corroborated by its strong, highly significant load-pH and load-conductivity main effects. The step-yield response has no active factor (F-test $p = 0.59$) and is reported as robust rather than modelled. Because the MVM model sets the step’s CQA, its coefficient table and ANOVA partition are given in full (Table 5.5, Table 5.6); the remaining coefficient tables are in Appendix C.

Table 5.4: Response-surface model adequacy across the four responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65
XMuLV LRF (log)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF (log)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

Table 5.5: Full quadratic-model coefficients for the MVM log-reduction (the CQA the step sets).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A ²	-0.322	0.199	-1.61	0.13	
B ²	0.149	0.199	0.747	0.469	
C ²	0.0224	0.199	0.112	0.912	
D ²	0.135	0.199	0.679	0.509	

The MVM coefficient table confirms the screening reading: the load-pH main effect is the largest term and the load-conductivity main effect the next, both highly significant, with no significant interaction or curvature term — a nearly planar surface over the characterized region. The ANOVA partition (Table 5.6) shows the model sum of squares dominating the residual, and the lack-of-fit component is not significant against pure error ($F = 0.88$, $p = 0.62$), so the quadratic model is an adequate description of the MVM response and no higher-order structure is indicated.

Table 5.6: ANOVA partition for the MVM response-surface model, with lack-of-fit tested against pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	11.7	14	0.834	8.14	0.000269
Residual	1.33	13	0.102	nan	nan
Lack of fit	0.994	10	0.0994	0.881	0.619
Pure error	0.339	3	0.113	nan	nan

The response surfaces over the two governing factors (Figure 5.1) show the pool HCP descending toward the high-pH / low-wash-conductivity corner and both viral log-reductions rising toward it, so the impurity- and virus-clearance optima coincide and there is no trade-off between them within the studied ranges. The model diagnostics for the MVM response (Figure 5.2) show residuals without structure against the predicted value, points on the normal reference line, and actual-versus-predicted scatter on the identity line, supporting the adequacy statement above.

Response-surface predictions (remaining factors held at set-point)

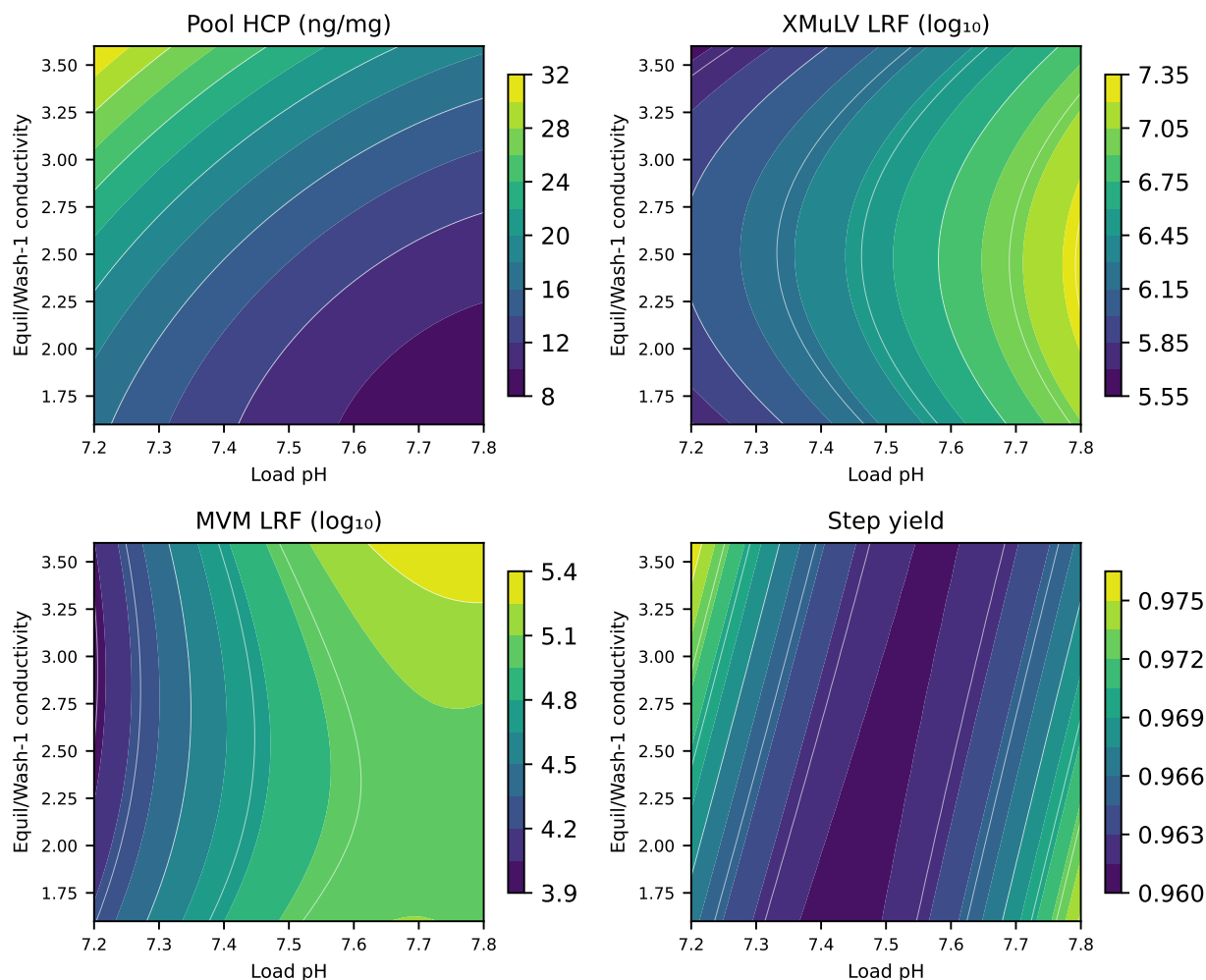


Figure 5.1: Response-surface predictions over load pH and equilibration/wash-1 conductivity (load conductivity and protein load held at set-point), for pool HCP, XMuLV and MVM log-reduction and step yield.

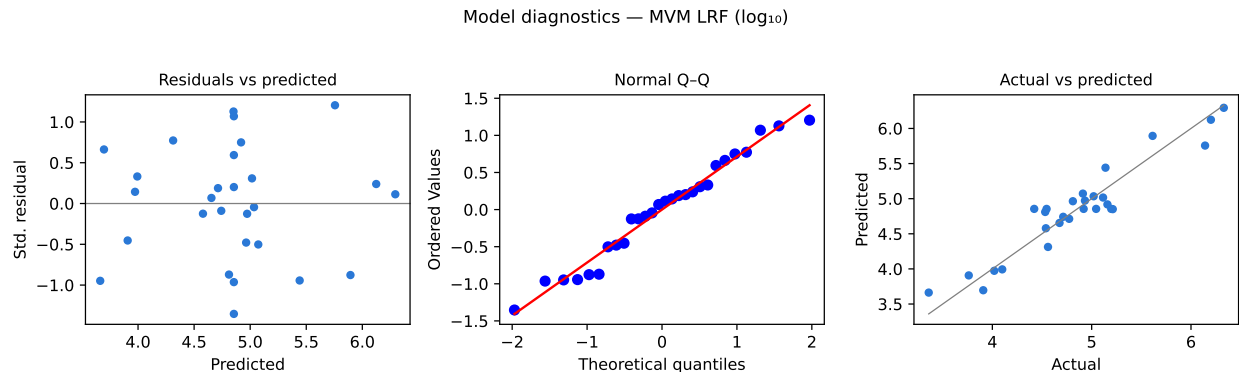


Figure 5.2: Model diagnostics for the MVM response-surface model: standardized residuals versus predicted, normal quantile–quantile, and actual versus predicted.

5.4 Mechanistic interpretation

The surfaces look as they do because a single physical mechanism governs the step. At the operating pH the antibody carries a slight net positive charge and transmits, while the more acidic host-cell protein, DNA, leached Protein A and model viruses carry net negative charge and bind the positively charged resin; the strength of that binding, and therefore the clearance, is set by the electrostatic environment. Raising the load pH increases the net negative charge of the impurities and virus and strengthens their retention, so the pool HCP falls and the XMuLV and MVM log-reductions rise; raising the wash-1 or the load conductivity screens the charge interaction and weakens retention, so the pool HCP rises and the viral log-reductions fall. The significant load-pH $\times$ wash-1-conductivity interaction on HCP is the expected coupling of these two levers — the conductivity penalty is largest where the pH is lowest and the binding already weakest. This is precisely why the impurity and viral load are assured over a multivariate operating region rather than by independent one-dimensional ranges: the acceptable value of any one factor depends on where the others sit, and only a joint region captures that dependence. The absence of a protein-load effect on clearance, over a range extending well above the set-point loading, further indicates that the binding capacity for impurities and virus is not exhausted at the intended load and that the load-related risk lies in the charge-variant quality of the feed rather than in a load set-point.

6 Design space

The anion-exchange design space is a multivariate region in the four well-controlled critical process parameters — load pH, equilibration/wash-1 conductivity, load conductivity and protein load — over which the governed attributes remain within acceptance. Its principal planes are the two the mechanism dictates: the load-pH $\times$ wash-1-conductivity plane that governs pool HCP, and the load-pH $\times$ load-conductivity plane that governs the XMuLV and MVM log-reductions. Within the characterized ranges the impurity and viral optima coincide toward the high-pH, low-conductivity corner, so the region has no interior trade-off; its binding constraints are the corners at which a viral log-reduction is lowest — low load pH combined with high load conductivity — which is the worst case the region must still satisfy, and at which the MVM contribution remains above its back-calculated step floor of 3.28 log .

The normal operating range of each parameter lies well inside this characterized region, which in turn lies inside the broader knowledge space bounded by the proven acceptable ranges of Section 7: load pH is operated at 7.35–7.65 within a characterized 7.2–7.8, wash-1 conductivity at 2.1–3.1 within 1.6–3.6 mS/cm, load conductivity at 4–7 within 3–8 mS/cm, and protein load at 120–280 within 50–300 g/L resin. Movement within this operating region is not a change and requires no assessment, whereas movement outside it would be evaluated under the quality system (CMC Biotech Working Group 2009; International Council for Harmonisation 2009). The claim is bounded to the characterized factor ranges and to the fitted response-surface models, and rests on the qualification of the scale-down model and its small-scale viral-spiking model (Section 5); it is not extrapolated beyond the studied edges.

7 Proven acceptable ranges

The proven acceptable range (PAR) of each parameter complements the multivariate design space with a per-attribute, per-parameter view: the sub-range of a parameter’s characterization range over which a given CQA stays within its acceptance criterion. The acceptance basis differs by attribute type and is returned by the model rather than assumed. For pool HCP the criterion is the drug-substance specification (< 100 ng/mg); for the viral-clearance attributes the criterion is the step-level required log-reduction, back-calculated from the cumulative requirement minus the clearance credited to the other steps — $3.28 \log$ for MVM and $3.78 \log$ for XMuLV — reflecting the modular nature of the viral-safety claim. Each parameter is reported with two proven acceptable ranges (Table 7.1): the at-set-point PAR, holding the other factors at their set-points, and the NOR-propagated PAR, in which the other factors vary within their normal operating ranges by Monte-Carlo of the fitted response-surface model — the reproducible, more conservative robustness criterion.

Table 7.1: Proven acceptable ranges for each governed CQA $\times$ parameter — characterization range and both the at-set-point and NOR-propagated PARs.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
Pool HCP (ng/mg)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
Pool HCP (ng/mg)	Load conductivity	3–8	mS/cm	3–8	3–8
Pool HCP (ng/mg)	Protein load	50–300	g/L resin	50–300	50–300
XMuLV LRF (log)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
XMuLV LRF (log)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
XMuLV LRF (log)	Load conductivity	3–8	mS/cm	3–8	3–8
XMuLV LRF (log)	Protein load	50–300	g/L resin	50–300	50–300
MVM LRF (log)	Load pH	7.2–7.8	pH	7.2–7.8	7.2–7.8
MVM LRF (log)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	1.6–3.6	1.6–3.6
MVM LRF (log)	Load conductivity	3–8	mS/cm	3–8	3–8

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
MVM LRF (log)	Protein load	50–300	g/L resin	50–300	50–300

Across every governed CQA and every multivariate parameter the two proven acceptable ranges coincide with the full characterization range (Table 7.1): the at-set-point and NOR-propagated PARs each span the entire studied range, for pool HCP, XMuLV and MVM alike. The interpretation is strong and specific. Each parameter’s whole characterization range is proven acceptable for every governed attribute, and remains acceptable even when the other factors co-vary within their normal operating ranges — so no single parameter, moved alone or with the rest of the operating point drifting within the NOR, drives any attribute out of specification anywhere in the studied space. The representative PAR plots make this visible for each governed CQA and its governing factor (Figure 7.1, Figure 7.2, Figure 7.3): the acceptable region, shaded, covers the full parameter axis in both the at-set-point and the NOR-propagated panel, with the set-point and normal operating range marked well inside it.

The relationship to the design space is therefore explicit. The proven acceptable ranges are univariate, NOR-robust statements about one parameter at a time; the design space is the multivariate region that captures the joint dependence among the factors. Because each univariate PAR already spans the full characterization range, the binding constraint on the operating region is not any single-parameter PAR but the multivariate worst-case corner identified in Section 6 — the low-pH, high-conductivity corner at which the viral log-reduction is lowest. This conclusion is bounded to the characterized ranges and the fitted models: it states that no univariate excursion within the studied space breaches acceptance, not that the parameters may be moved without limit.

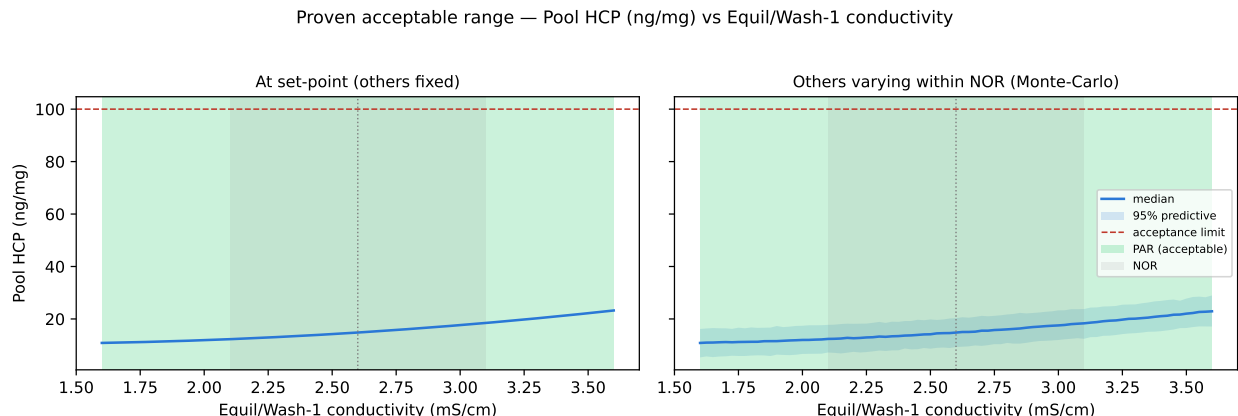


Figure 7.1: Proven acceptable range for pool HCP versus its governing factor (equilibration/wash-1 conductivity): at set-point (left) and with the other factors varying within their NOR (right); acceptable region shaded, set-point and NOR marked.

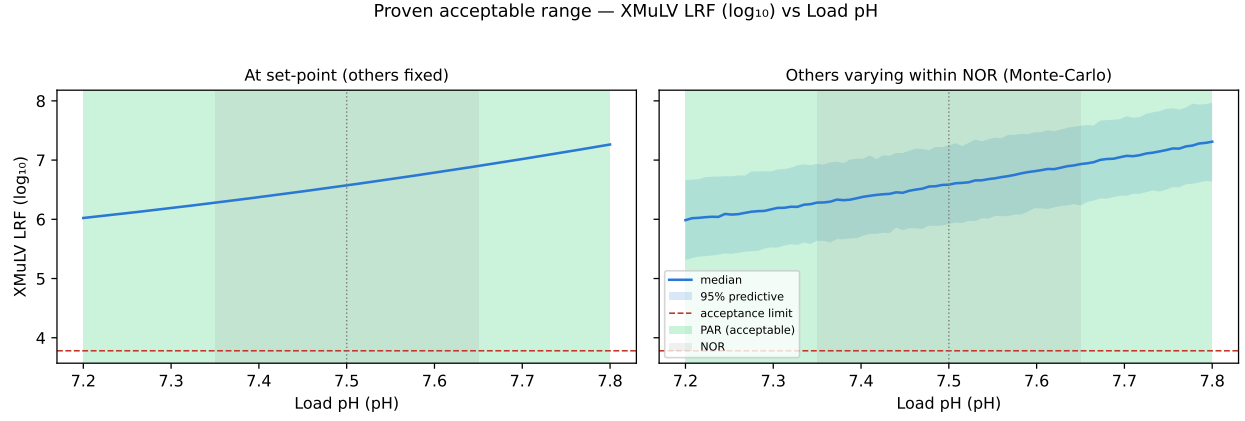


Figure 7.2: Proven acceptable range for XMuLV log-reduction versus its governing factor (load pH), at set-point and NOR-propagated.

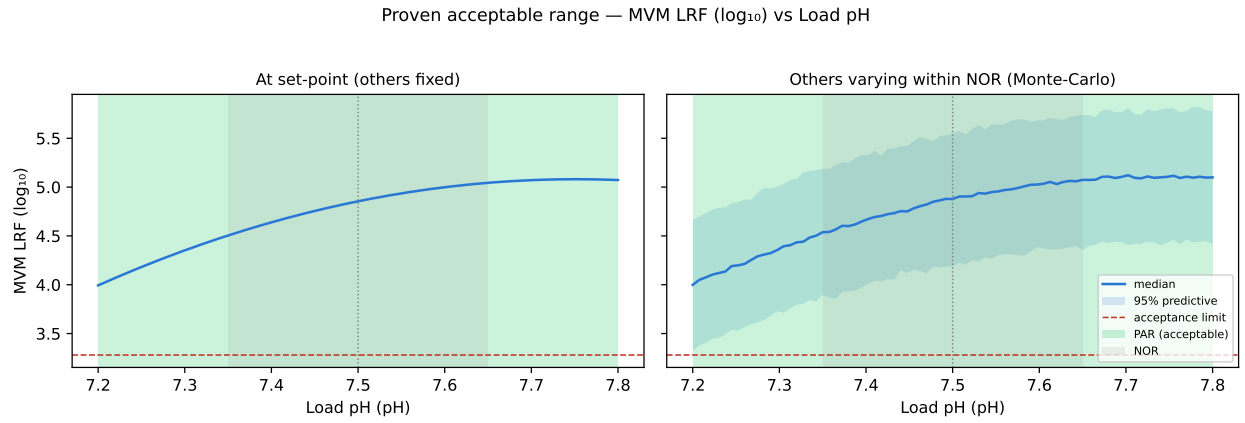


Figure 7.3: Proven acceptable range for MVM log-reduction versus its governing factor (load pH), at set-point and NOR-propagated.

8 Process capability and robustness

The commercial-scale capability of every governed attribute meets its acceptance criterion with margin (Table 8.1), estimated by Monte-Carlo simulation of 2,000 batches with each parameter drawn across its normal operating range and propagated through the fitted response-surface models. The MVM clearance — the CQA the step sets — is the tightest capability of the entire drug substance: its one-sided Cpk of 1.51 is the process minimum (1.51), yet the simulated cumulative MVM clearance ($10.4 \pm 0.4 \log$) sits above its $8.6 \log$ requirement, to which anion exchange contributes its $4.71 \log$ as the first and principal MVM-clearance step alongside virus filtration (PCR-009). The XMuLV clearance is comparably capable (Cpk = 1.53, cumulative $19.5 \log$ against a $16.7 \log$ requirement), to which this step contributes $5.95 \log$ orthogonally to low-pH inactivation (PCR-006) and virus filtration. The pool HCP is highly capable (Cpk = 6.14, mean 15.2 ng/mg against a 100 ng/mg limit), with residual DNA and leached Protein A far inside their limits (Cpk = 10.8 and 2785), the latter cleared here after its principal removal at Protein A capture (PCR-005) and cation exchange (PCR-007). The two viral capabilities are the binding ones; the impurity capabilities carry ample margin. These estimates are bounded to the normal operating ranges and to the scale-down and small-scale spiking models, and the cumulative viral figures are consolidated across the three orthogonal viral steps in PCMR-001, not claimed by this step alone.

Table 8.1: Commercial-scale capability of the attributes governed by anion exchange.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	$2.79\text{e}+03$
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

9 Parameter classification

Each parameter is classified from its demonstrated effect on the governed attributes and the risk it carries within the operating region; the classifications are summarized in Table 4.1 and justified below.

- **Load pH — WC-CPP.** The dominant lever on both viral log-reductions and, with wash-1 conductivity, on pool HCP; it is therefore controlled as a well-controlled critical process parameter, at 7.5 within a 7.35–7.65 normal operating range. Its whole characterization range is a proven acceptable range for every governed CQA (Section 7), so the residual risk is fully managed by the routine pH control the NOR represents.
- **Equilibration/Wash-1 conductivity — WC-CPP.** The governing factor for pool HCP and, with load pH, a determinant of impurity clearance; controlled as a WC-CPP at 2.6 mS/cm within 2.1–3.1. Its effect is significant but its proven acceptable range spans the full studied range, so the attribute stays in specification across the NOR.
- **Load conductivity — WC-CPP.** The secondary lever on the viral log-reductions, acting in the opposite direction to load pH; controlled as a WC-CPP at 5.5 mS/cm within 4–7. It defines, with load pH, the worst-case corner of the viral design space, which is why it is controlled rather than treated as a general parameter despite its smaller effect.
- **Protein load — WC-CPP.** Carries a credible risk to the impurity and viral load and is therefore controlled as a WC-CPP; in representative-quality load material it showed no significant effect on clearance or recovery over a range extending well above the set-point, so its acceptable range is wide. The characterization localized the load-related risk to the charge-variant quality of the load material (Section 13), which is controlled as a feed-input attribute rather than by a load set-point.
- **Operating flow rate — WC-CPP.** Assessed univariately and found to have no significant effect on clearance or recovery over 150–450 cm/hr; it is nonetheless controlled as a WC-CPP at 300 cm/hr within 200–400, because residence time carries a credible mechanistic risk to a binding-limited clearance step and a wide, evidenced acceptable range is the appropriate control rather than declassification.

No parameter at this step required the most stringent (full CPP) control, and the step carries no key process parameter: all five parameters are well-controlled critical process parameters, each with a demonstrated effect and an evidenced acceptable range.

10 Contribution to the control strategy

Anion exchange contributes to the control strategy as the process's principal parvovirus- clearance step and as a polishing step for enveloped virus and process impurities. Its contributions are: control of the four multivariate WC-CPPs and the flow rate to the normal operating ranges of Table 4.1; in-process monitoring of the flow-through-pool HCP by ELISA (AMV-3012) and UV-triggered pool collection at the corrected set-point established under Section 13; verification of the load-material charge-variant quality by icIEF (AMV-3013) as a feed-input control, added as a direct consequence of the load-material deviation; and the resin life-cycle, packing and sanitization controls of SOP-2008. The MVM clearance the step sets is credited toward the cumulative viral-safety claim modularly, alongside the orthogonal contributions of low-pH inactivation (PCR-006) and virus filtration (PCR-009); the enveloped-virus and impurity clearances are likewise partial contributions to attributes also controlled upstream at Protein A capture (PCR-005) and cation exchange (PCR-007). The step does not, by itself, guarantee the cumulative viral-safety claim or the release-level impurity attributes: its role is to provide an orthogonal, robust clearance increment whose consolidation with the other steps is performed in the master report PCMR-001.

11 Discussion

The characterization establishes that the anion-exchange step is well understood and robust across the studied ranges, and that its behaviour is the behaviour prior platform knowledge predicted. A single electrostatic mechanism accounts for every governed response: load pH and conductivity set the strength of impurity and virus binding, the impurity and viral optima coincide, and the step is high-recovery and insensitive to protein load and flow rate over wide ranges. The response-surface models are adequate over the characterized region and, for pool HCP and XMuLV, additionally predictive; the more modest predicted R^2 of the MVM model is attributable to the narrower dynamic range of the parvovirus spiking assay against its reproducibility, not to model inadequacy, and the MVM classification and design space rest on its strong, highly significant main effects and its non-significant lack of fit. Confidence in these conclusions rests on the qualification of the scale-down model, whose input and output attributes matched at-scale development data, and on the qualified small-scale viral-spiking model.

Three deviations were recorded, and their treatment is summarized adverse-first in Section 13. The most consequential was a non-representative, deamidated load in the first DoE execution, which — through an anomalous protein-load $\times$ wash-1-conductivity interaction — drove the flow-through pool out of its in-process limit at the high-load/high-conductivity corner; because an operating region cannot be defined on a non-representative load, the affected designs were invalidated and re-executed in full on a requalified load, and the requalified execution is the analysis reported throughout this document. The two remaining deviations — a permissive UV pool-collection set-point common to both executions, and a single wash-buffer lot released slightly below its pH range — were bounded by argument and, respectively, corrected by modelling and verification runs and retained. The principal limitations of the study are those inherent to Stage-1 characterization: the reliance on a scale-down model, the use of a small-scale viral-spiking model whose log-reduction range is narrower than the cumulative claim it supports, and the assessment of resin performance over a limited number of cycles rather than the full life-cycle. Each is managed forward — the scale-down and spiking models by their qualification and by confirmation at scale, and the resin life-cycle by the controls of SOP-2008 and by continued process verification. No parameter required control more stringent than the well-controlled critical designation, which is the appropriate outcome for a step whose attributes remain in specification across their full proven acceptable ranges.

12 Conclusions

The anion-exchange polishing step is well characterized, well understood and robust across the ranges studied. A multivariate operating region in the four well-controlled critical process parameters was defined over which the governed attributes remain within acceptance; the proven acceptable range of each parameter spans its full characterization range and is robust to NOR co-variation, so the binding constraint on the operating region is the multivariate worst-case corner rather than any univariate range. Every governed attribute meets its commercial-scale acceptance criterion with margin, the tightest being the MVM clearance the step sets ($Cpk = 1.51$, the process minimum), which anion exchange establishes as the first and principal parvovirus-clearance step alongside virus filtration. The three recorded deviations were investigated, impact-assessed and resolved without altering the parameter classifications or the operating region. The step does not by itself deliver the cumulative viral-safety claim; its contribution is consolidated with the other viral and polishing steps in the master report PCMR-001, into which the classifications, operating region and control contributions defined here are carried forward.

13 Deviations from the plan

Three deviations were recorded during execution; each was investigated, its root cause established, and its impact on the reported conclusions assessed (Table 13.1). One invalidated the first DoE execution and forced a full re-execution, one was a set-point correction common to both executions, and one was a single out-of-range buffer lot retained on a bounded-impact argument. None altered the parameter classifications or the operating region defined above.

Table 13.1: Recorded deviations for the anion-exchange characterization.

Devi- ation	Summary	Detected during	Disposition
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated_and_re_executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected_by_modelling_and_verification_runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained

13.1 DEV-008-01 — Non-representative (deamidated) load; designs invalidated and re-executed

In the first execution of the screening and response-surface designs, the anion-exchange load — the cation-exchange eluate — was subsequently found to carry an elevated acidic charge-variant (deamidation) burden of 33.5% by icIEF (AMV-3013), against 21% for representative material, the result of an out-of-trend extended neutralized hold of 36 hours (lot LOT-CEX-8842) that promoted asparagine deamidation. In that first execution, at the high-protein-load / high-wash-1-conductivity corner of the design, the flow-through-pool HCP rose sharply above its in-process action limit and the pool purity became unacceptable, through an anomalous protein-load $\times$ wash-1-conductivity interaction that is absent from representative material. In the invalidated data the protein-load main effect and its interaction with wash-1 conductivity are both large and significant (effects 36 and 35 ng/mg, p 0.002; Table 13.2). Because an acceptable operating region cannot be defined on a non-representative load, the affected screening and response-surface designs were invalidated for operating-region definition and re-executed in full on a requalified, representative load.

The root cause is confirmed by the requalified data. In the requalified-load design reported throughout this document, the protein-load $\times$ wash-1-conductivity interaction is statistically absent (effect 0.22 ng/mg, $p = 0.86$; Table 5.3) and the protein-load main effect is likewise non-significant. The anomalous interaction seen in the first execution was therefore an artifact of the degraded (deamidated) load — whose additional acidic charge variants competed for the resin and coupled the protein load to the wash conductivity — and not an intrinsic property of the step. The disposition is that the superseded dataset is retained and referenced for this root-cause confirmation only; the reported analysis, classifications and operating region derive entirely from the requalified execution. The deviation also generated a standing feed-input control: the load-material charge-variant quality is verified by icIEF before the step (Section 10).

Table 13.2: Pool-HCP screening effects from the invalidated first execution (superseded; shown for root-cause confirmation only).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	50.4	25.2	3.89	6.46	0.000195	***
D	35.6	17.8	3.89	4.57	0.00181	**
BD	34.8	17.4	3.89	4.46	0.00211	**
A	-22.7	-11.4	3.89	-2.92	0.0194	*
AB	-14.6	-7.3	3.89	-1.87	0.0978	
C	-10.2	-5.1	3.89	-1.31	0.227	
BC	-9.11	-4.56	3.89	-1.17	0.276	
AD	-8.62	-4.31	3.89	-1.11	0.301	
CD	-6.46	-3.23	3.89	-0.83	0.431	
AC	5.25	2.62	3.89	0.674	0.519	

13.2 DEV-008-02 — Permissive trailing-edge UV pool-collection set-point

In both executions the trailing-edge UV set-point that stops pool collection was set slightly permissive, at 180 mAU rather than the intended 120 mAU, and was detected on post-execution pool-fraction reconciliation against the HCP assay (AMV-3012). Because the identical set-point was applied to every run, its effect is a common-mode offset: it shifts the absolute flow-through-pool HCP level by an approximately constant amount but does not change the factor effects, the two-factor interactions or the geometry of the operating region, since adding a constant to every response changes only the model intercept and not the coefficients. The parameter classifications and the design-space geometry reported here are therefore unaffected; only the absolute in-process pool-collection set-point required correction. The set-point was corrected to 120 mAU and the correction verified by 3 confirmation runs at the corrected set-point plus 2 robustness runs, which bounded the corrected pool-HCP level with a relative 95% confidence half-width of 23.6%. The corrected set-point is the one carried into the control strategy (Section 10).

13.3 DEV-008-03 — Equilibration/Wash-1 buffer released below its pH range

On one screening run the equilibration/wash-1 buffer lot (LOT-BUF-8815) was released at a measured pH of 7.32, below its 7.4 lower acceptance limit and its 7.5 target, detected on buffer release testing per SOP-2103; the root cause was a titration-endpoint error at buffer preparation. The deviation was retained. Flow-through clearance at this step is governed by the load pH and the wash-1 conductivity, not by the wash-buffer pH per se, and the small buffer-pH offset falls well within the load-pH range the study characterized, so the affected run remains representative of the operating region and does not alter the fitted effects. The impact assessment therefore brackets the excursion within the studied space and confirms that no re-execution of the run was required.

Appendix A — Screening design matrix and responses

The complete coded screening design and its measured responses are given in Table 13.3; the factor letters follow Table 4.2, and centre points are marked in the type column.

Table 13.3: Coded screening design matrix and responses (requalified execution).

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
1	factorial	-1	-1	-1	-1	12.5	6.16	4.59	0.99
2	factorial	-1	-1	-1	1	15.6	6.15	4.43	0.975
3	factorial	-1	-1	1	-1	12.1	5.47	3.76	0.951
4	factorial	-1	-1	1	1	14.6	5.56	3.96	0.964
5	factorial	-1	1	-1	-1	35.7	6.34	4.74	0.965
6	factorial	-1	1	-1	1	35	6.3	4.83	0.995
7	factorial	-1	1	1	-1	29.2	5.92	4.01	0.974
8	factorial	-1	1	1	1	29.8	5.95	3.94	0.969
9	factorial	1	-1	-1	-1	9.62	7	5.91	0.973
10	factorial	1	-1	-1	1	7.37	7.4	5.61	0.953
11	factorial	1	-1	1	-1	8.69	6.61	5.25	0.972
12	factorial	1	-1	1	1	8.04	6.71	4.89	0.973
13	factorial	1	1	-1	-1	12.7	6.59	5.91	0.995
14	factorial	1	1	-1	1	15	8.34	5.97	0.981
15	factorial	1	1	1	-1	12.9	6.69	5.15	0.966

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
16	facto- rial	1	1	1	1	15	6.52	5.31	0.965
17	center	0	0	0	0	16.2	6.42	4.54	0.966
18	center	0	0	0	0	17.5	6.8	5.08	0.975
19	center	0	0	0	0	11.6	5.95	4.83	0.974

Appendix B — Response-surface design matrix and responses

The complete coded face-centred central-composite design and its responses are given in Table 13.4.

Table 13.4: Coded response-surface design matrix and responses (requalified execution).

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
1	factorial	-1	-1	-1	-1	13.5	6.72	4.55	0.951
2	factorial	-1	-1	-1	1	17.3	6.03	5.21	0.989
3	factorial	-1	-1	1	-1	14.1	5.87	4.02	0.995
4	factorial	-1	-1	1	1	15	5.3	3.76	0.951
5	factorial	-1	1	-1	-1	28.5	6.53	4.77	0.988
6	factorial	-1	1	-1	1	25.5	6.71	4.71	0.963
7	factorial	-1	1	1	-1	31.3	5.36	3.91	0.995
8	factorial	-1	1	1	1	32.3	5.06	3.36	0.966
9	factorial	1	-1	-1	-1	5.66	7.5	6.14	0.974
10	factorial	1	-1	-1	1	8.37	7.83	5.61	0.989
11	factorial	1	-1	1	-1	7.56	6.49	4.54	0.972
12	factorial	1	-1	1	1	7.43	7.13	4.68	0.988
13	factorial	1	1	-1	-1	17.6	7.18	6.2	0.974
14	factorial	1	1	-1	1	18	8.12	6.33	0.968
15	factorial	1	1	1	-1	13.9	6.58	4.53	0.96
16	factorial	1	1	1	1	13.9	6.37	5.16	0.973

Run	Type	A	B	C	D	Pool HCP (ng/mg)	XMuLV LRF (log)	MVM LRF (log)	Step yield
17	axial	-1	0	0	0	23	5.83	4.1	0.971
18	axial	1	0	0	0	12	7.37	4.91	0.965
19	axial	0	-1	0	0	8.1	6.42	4.93	0.946
20	axial	0	1	0	0	27.5	6	5.02	0.972
21	axial	0	0	-1	0	15.6	7.44	5.14	0.952
22	axial	0	0	1	0	15.5	5.93	4.56	0.967
23	axial	0	0	0	-1	13	6.52	4.81	0.968
24	axial	0	0	0	1	14.3	6.7	5.11	0.952
25	center	0	0	0	0	15	6.59	4.42	0.958
26	center	0	0	0	0	10.7	6.03	5.2	0.975
27	center	0	0	0	0	16.3	7.05	5.04	0.977
28	center	0	0	0	0	12.6	6.86	4.92	0.953

Appendix C — Full effect and coefficient tables

The full screening effect estimates for the XMuLV, MVM and step-yield responses are given in Table 13.5, Table 13.6 and Table 13.7, and the full response-surface coefficient tables for pool HCP, XMuLV and step yield — those not reproduced in Section 5 — in Table 13.8, Table 13.9 and Table 13.10.

Table 13.5: Screening effect estimates — XMuLV log-reduction.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

Table 13.6: Screening effect estimates — MVM log-reduction.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Table 13.7: Screening effect estimates — step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

Table 13.8: Response-surface coefficients — pool HCP.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A ²	1.91	1.65	1.16	0.268	
B ²	2.22	1.65	1.34	0.202	
C ²	-0.0187	1.65	-0.0113	0.991	
D ²	-1.98	1.65	-1.2	0.253	

Table 13.9: Response-surface coefficients — XMuLV log-reduction.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	
CD	-0.0751	0.0785	-0.957	0.356	

Term	Coef.	Std. err.	t	p-value	Sig.
A ²	0.0696	0.195	0.356	0.727	
B ²	-0.328	0.195	-1.68	0.118	
C ²	0.152	0.195	0.779	0.45	
D ²	0.0761	0.195	0.39	0.703	

Table 13.10: Response-surface coefficients — step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A ²	0.00932	0.00902	1.03	0.32	
B ²	0.000524	0.00902	0.0581	0.955	
C ²	0.00152	0.00902	0.168	0.869	
D ²	0.00177	0.00902	0.196	0.847	

Appendix D — Analytical methods summary

The validated analytical methods that measured the step's responses and load-material quality are summarized in Table 13.11; method performance and reportable ranges are held in the respective validation reports.

Table 13.11: Analytical methods supporting the anion-exchange characterization.

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID ₅₀)	Method validation
AMV-3018	MVM Infectivity Titre (TCID ₅₀ /qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.
- International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.